

Dynamic Pricing Strategy

Business Recommendations

Cult Fitness Class Booking Analysis

Dataset records analyzed: 3,271

1. Business Objective

- Use booking demand and class capacity to adjust prices dynamically.
- Increase utilization during low-demand periods while capturing additional value during high-demand periods.
- Make pricing responsive to occupancy, time, class popularity and location.
- Use historical booking data as the foundation for a repeatable pricing process.

2. Key Business Signals

- Occupancy = Number Booked $\div$ Maximum Capacity.
- Low occupancy indicates available capacity and an opportunity for targeted discounts.
- High occupancy indicates strong demand and potential for a controlled price increase.
- Activity-level and location-level booking patterns can identify consistently popular segments.
- Peak-hour analysis can support time-based pricing.

3. Recommended Dynamic Pricing Rules

- Occupancy below 40%: reduce price by about 10% to stimulate demand.
- Occupancy between 40% and 70%: maintain the standard price.
- Occupancy above 70%: increase price by about 10% to capture stronger demand.
- Peak-hour classes can receive a modest additional adjustment, such as 5%, subject to business testing.
- Apply price caps/floors so prices remain within acceptable customer and business limits.

4. Business Recommendation — Low-Demand Classes

- Use targeted discounts rather than broad price reductions.
- Prioritize classes with repeatedly low occupancy.
- Promote low-demand sessions through offers, bundles or off-peak memberships.
- Measure whether lower prices actually generate incremental bookings before expanding the discount.

5. Business Recommendation — High-Demand Classes

- Use controlled price increases for consistently high-occupancy classes.
- Avoid aggressive increases that could reduce customer retention or perceived value.
- Monitor waitlists, cancellations and booking speed alongside occupancy.
- Consider protecting a portion of capacity for members or loyal customers.

6. Time & Location Strategy

- Identify peak booking windows and compare occupancy across hours.
- Use time-based adjustments only where historical demand supports them.
- Compare locations using average bookings and occupancy rather than raw booking counts alone.
- Test different pricing rules by site because demand patterns can differ by location.

7. Class Popularity Strategy

- Calculate average bookings by Activity.
- Identify activities with consistently above-average demand.
- Use popularity as a supporting signal, not the only pricing factor.
- Review activity-level performance regularly because popularity can change over time.

8. Simulation & Measurement

- Back-test the pricing rules on historical records before deploying them.
- Track original price, simulated price, bookings, occupancy and revenue.
- Do not assume bookings stay constant after a price change; incorporate the price-elasticity model for a more realistic simulation.
- Evaluate MAE or other forecasting metrics for demand forecasts and compare simulated revenue against the baseline.

9. Recommended KPIs

- Occupancy Rate — how much class capacity is being used.
- Revenue per Class — revenue generated by each session.
- Revenue per Available Seat — revenue relative to available capacity.
- Booking Conversion / Booking Pace — how quickly seats are filled.
- Cancellation Rate and Waitlist Size — customer response to pricing and capacity.
- Average Price and Discount Rate — monitor pricing changes and customer exposure.

10. Implementation Roadmap

- Phase 1: Establish baseline KPIs and clean historical booking data.
- Phase 2: Validate demand forecasts and price elasticity.
- Phase 3: Back-test dynamic pricing rules using historical data.
- Phase 4: Run a controlled pilot on selected classes or locations.
- Phase 5: Compare pilot results with a control group and refine thresholds.
- Phase 6: Automate price recommendations and create a monitoring dashboard.

11. Dataset Snapshot

Metric	Value
Records	3,271
Average Occupancy	56.5%
Current Revenue (historical)	99,073,581.00
Simulated Revenue (simple rule)	101,849,875.70
Simulated Revenue Difference	2,776,294.70

Note: the simple revenue simulation assumes bookings remain unchanged after the price adjustment. A price-elasticity model should be used for a more realistic business estimate.

12. Final Business Recommendations

- Start with transparent, rule-based pricing that managers can audit.
- Use occupancy as the primary signal, supported by time, activity and location factors.
- Keep price changes moderate and introduce floors/caps.
- Validate every pricing rule with historical back-testing and controlled experiments.
- Connect dynamic pricing to the price-elasticity model so expected booking changes are included.
- Monitor customer experience KPIs as well as revenue; optimize for sustainable utilization, not revenue alone.

Conclusion

Dynamic pricing can turn historical booking data into an actionable decision system. The recommended approach is to start with simple occupancy-based rules, validate them using elasticity and forecasting models, pilot them in controlled settings, and continuously monitor business and customer KPIs.